

Module 1: Intelligent Agents & PEAS

Definitions of AI, Turing Test, Rationality, PEAS Framework & 5 Agent Types

Module I: Preliminaries & Intelligent Agents — Complete 7-Topic Syllabus Tracker

- Topic 1:** What is AI? (4 Classical Approaches & Rational Agent Paradigm)
- Topic 2:** Evolution of AI (1950 Turing Test to 2020s Generative AI)
- Topic 3:** Intelligent Agents (Sense-Think-Act Loop & PEAS Framework)
- Topic 4:** Concept of Rationality (Rationality vs. Omniscience)
- Topic 5:** Nature of Environments (O-D-E-D-C-N Taxonomy)
- Topic 6:** Structure of Agents (5 Architectural Agent Types)
- Topic 7:** Real-World Applications of AI (Healthcare, Robotics, Vision)

Topic 1 & 2: Definitions, Approaches & Historical Evolution of AI

Artificial Intelligence is the branch of computer science dedicated to designing computational systems that synthesize capabilities traditionally associated with human intelligence: reasoning, perception, learning, planning, and goal-directed decision making.

Approach	Philosophical / Methodological Basis	Standard Evaluation Benchmark
1. Thinking Humanly	Cognitive science modeling of the inner workings of human thought processes.	GPS (General Problem Solver), Cognitive Neuroimaging.
2. Acting Humanly	Empirical behavioral replication of human performance.	Turing Test (1950) by Alan Turing.
3. Thinking Rationally	Formal logic and laws of thought (Syllogisms, First-Order Logic).	Deductive Inference Engines and Proof Systems.
4. Acting Rationally	Rational Agent Approach (Modern Standard): Maximizing expected performance given available information.	Goal achievement and expected utility maximization.

 **Memory Hook: Evolution Timeline (T-D-E-ML-DL-GenAI)**
1950: Turing Test → **1956:** Dartmouth Conference (John McCarthy coining "AI") → **1980s:** Expert Systems (R1/XCON) → **1997:** Deep Blue defeats Kasparov → **2011:** IBM Watson wins Jeopardy → **2016:** AlphaGo defeats Lee Sedol → **2020s:** Large Multimodal Transformers & Generative AI.

Topic 3 & 4: Intelligent Agents, PEAS & The Concept of Rationality

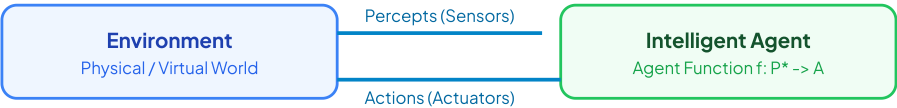


Figure 1.1: The Core Sense-Think-Act Intelligent Agent Interaction Loop

Rationality $\neq$ Omniscience

An **omniscient agent** knows the *actual outcome* of its actions in advance and acts with superhuman clairvoyance (impossible in reality). A **rational agent** chooses the action that maximizes its *expected performance measure*, given its accumulated percept history sequence P^* and built-in knowledge base!

PEAS Specification Framework (Autonomous Taxi Example):

P — Performance Measure: Safety, passenger comfort, speed, legal compliance, fuel economy, profit.

E — Environment: City streets, highways, pedestrians, weather conditions, other vehicular traffic.

A — Actuators: Steering wheel, accelerator, brake pedal, horn, turn signals, display screen.

S — Sensors: Video cameras, LiDAR, radar, ultrasonic sonar, GPS, speedometer, odometer, engine sensors.

Topic 5: Nature of Environments (O-D-E-D-C-N Taxonomy)

Environment Dimension	Core Conceptual Distinction	Representative Example
1. Fully vs. Partially Observable	Sensors detect the complete state of the world vs. state is obscured by noise or missing sensors.	Chess (Fully) vs. Poker / Driving (Partially)
2. Deterministic vs. Stochastic	Next state determined 100% by current state + action vs. randomness and uncertainty exist.	Crossword (Deterministic) vs. Weather / Taxi (Stochastic)
3. Episodic vs. Sequential	Each action episode is independent vs. current action alters future states and rewards.	Image Defect Sorting (Episodic) vs. Chess (Sequential)
4. Static vs. Dynamic	Environment remains unchanged while agent deliberates vs. changes continuously in real time.	Sudoku (Static) vs. Autonomous Driving (Dynamic)
5. Discrete vs. Continuous	Countable discrete states and actions vs. continuously varying physical quantities.	Tic-Tac-Toe (Discrete) vs. Robot Arm Control (Continuous)
6. Single-Agent vs. Multi-Agent	Agent operates alone vs. multiple agents interact cooperatively or competitively.	Crossword Puzzle (Single) vs. Chess / Stock Market (Multi)

Topic 6: The 5 Architectural Agent Types

Agent Type	Decision Mechanism & Internal State	Key Limitation / Advantage
1. Simple Reflex	Direct Condition-Action Rules ('IF condition THEN action') based strictly on current percept.	Zero memory; fails completely in partially observable environments.
2. Model-Based Reflex	Maintains internal state reflecting "how the world evolves" and "how actions affect the world".	Overcomes partial observability by tracking hidden state history.
3. Goal-Based	Combines internal state with explicit Goal information using Search & Planning algorithms.	Flexible: goals can be altered dynamically without rewriting rules.
4. Utility-Based	Evaluates state desirability using an explicit real-valued Utility Function $U(s)$.	Handles conflicting goals and probabilistic risk-reward tradeoffs.
5. Learning Agent	Decoupled into 4 Components : (1) Learning Element, (2) Performance Element, (3) Critic (evaluator), (4) Problem Generator (exploration).	Autonomously improves performance over time from raw environmental feedback.



M1 Active Recall & Exam Questions

Q1. Define an Intelligent Agent. Explain the 4 components of a Learning Agent with a neat architecture diagram. (10 Marks)

An **Intelligent Agent** is an autonomous entity that perceives its environment through sensors and takes actions using actuators to achieve its goals ($f : P^* \rightarrow A$).

The 4 Components of a Learning Agent:

1. *Learning Element*: Responsible for making improvements by modifying the agent's internal behavior rules.
2. *Performance Element*: Responsible for selecting external actions based on percepts (the operational agent).
3. *Critic*: Evaluates the agent's performance against an external performance standard and provides feedback to the learning element.
4. *Problem Generator*: Suggests experimental exploratory actions that lead to new, informative experiences rather than suboptimal repetition.